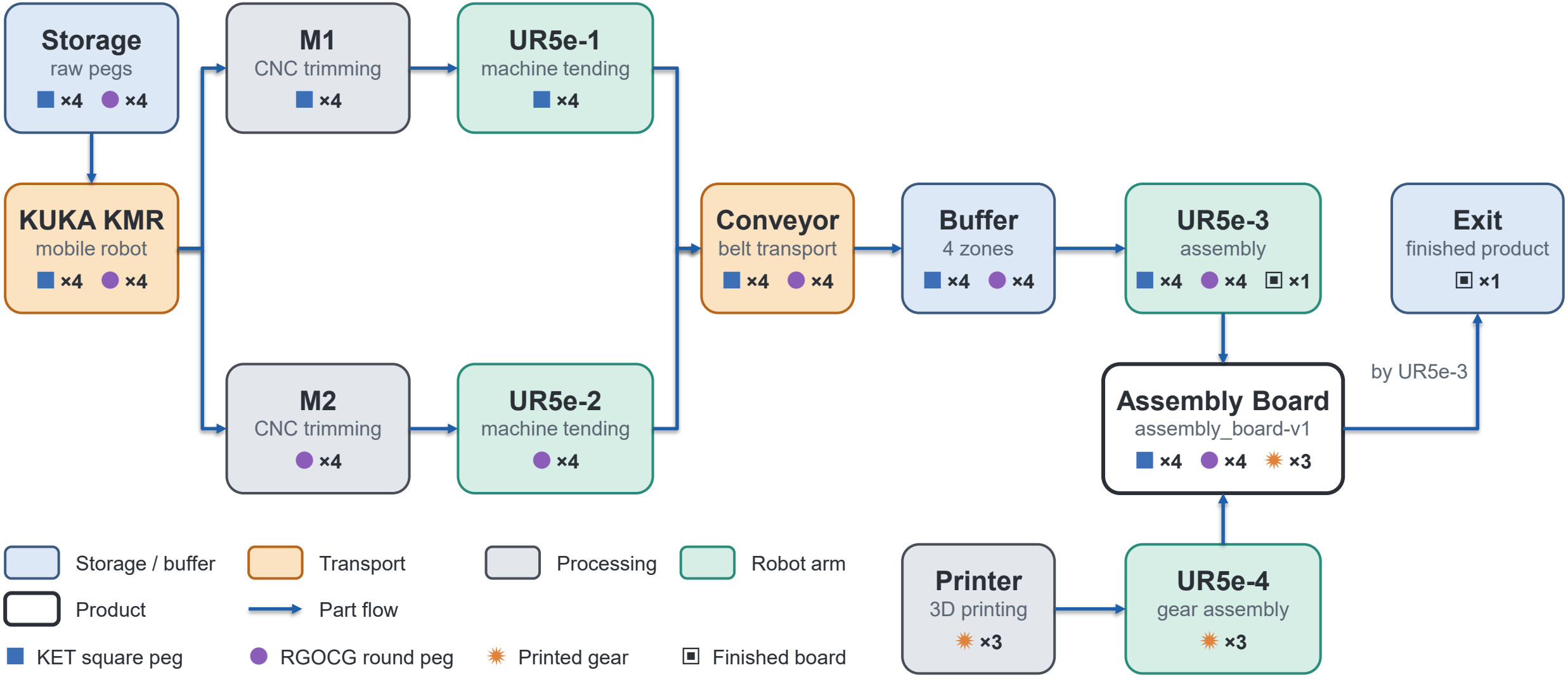
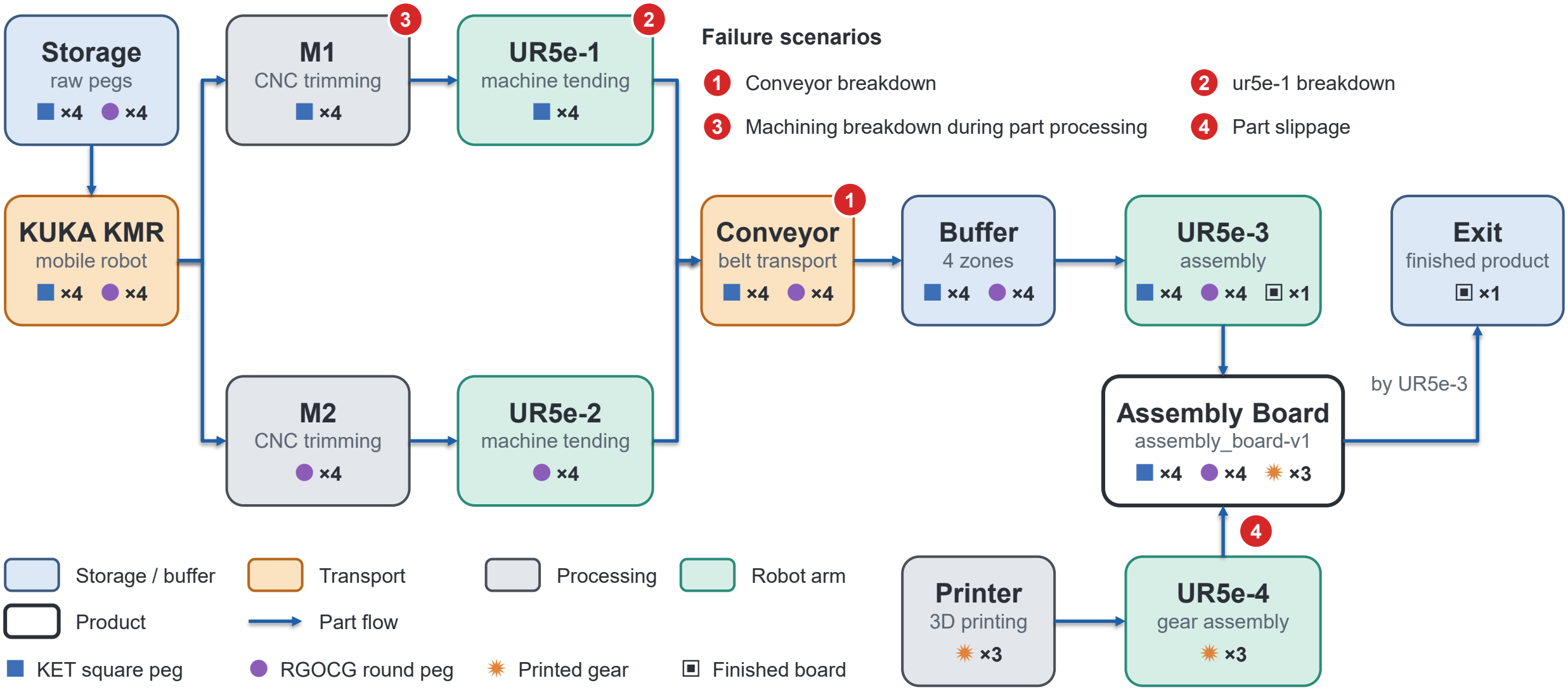
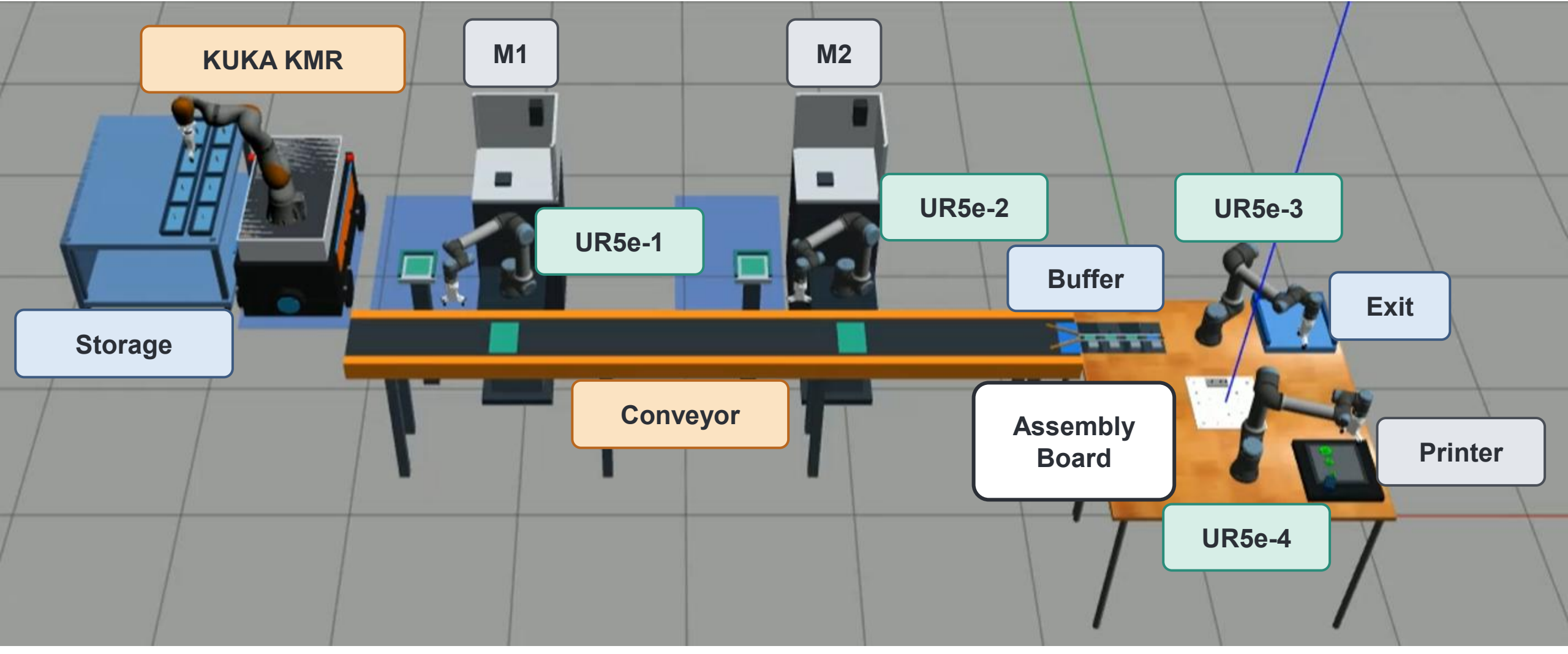

Storage
raw pegs
×4 ×4
KUKA KMR
mobile robot
×4 ×4
M1
CNC trimming
×4
M2
CNC trimming
×4
UR5e-1
machine tending
×4
UR5e-2
machine tending
×4
Conveyor
belt transport
×4 ×4
Buffer
4 zones
×4 ×4
UR5e-3
assembly
×4 ×4 ×1
Exit
finished product
×1
by UR5e-3
Assembly Board
assembly_board-v1
×4 ×4 ×3
Printer
3D printing
×3
UR5e-4
gear assembly
×3
Storage / buffer
Transport
Processing
Robot arm
Product
Part flow
KET square peg
RGOCG round peg
Printed gear
Finished board


Storage
raw pegs
×4 ×4
KUKA KMR
mobile robot
×4 ×4
M1
CNC trimming
×4
3
M2
CNC trimming
×4
UR5e-1
machine tending
×4
2
UR5e-2
machine tending
×4
Conveyor
belt transport
×4 ×4
1
Buffer
4 zones
×4 ×4
UR5e-3
assembly
×4 ×4 ×1
Exit
finished product
×1
by UR5e-3
Assembly Board
assembly_board-v1
×4 ×4 ×3
4
Printer
3D printing
×3
UR5e-4
gear assembly
×3
Failure scenarios
1 Conveyor breakdown
2 ur5e-1 breakdown
3 Machining breakdown during part processing
4 Part slippage
Storage / buffer
Transport
Processing
Robot arm
Product
Part flow
KET square peg
RGOCG round peg
Printed gear
Finished board


KUKA KMR
M1
M2
UR5e-2
UR5e-3
UR5e-1
Buffer
Exit
Storage
Conveyor
Assembly Board
Printer
UR5e-4